Wentao Zhang
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14 May 2026

Selection Committee
School of Electrical Engineering and Computer Science
The University of Queensland, St Lucia QLD 4072, Australia

Dear Selection Committee,

Application for Lecturer in Electrical Engineering / Biomedical Engineering, R-63679

I am writing to apply for the Lecturer position in Electrical Engineering / Biomedical Engineering (**R-63679**). I am currently a Research Assistant at the **Power Electronics and Machines Centre (PEMC), University of Nottingham**, having completed my **PhD in Electrical Engineering at Southeast University, Nanjing, in September 2025**. The combination of broad teaching across the electrical engineering curriculum and a research home in a School with strong **Imaging, Sensing and Biomedical Engineering** activity is an excellent fit for the methodology-led programme I want to build at UQ.

My research develops **multiphysics modelling and AI-driven surrogate techniques** for the analysis, design and optimisation of magnetic and electromagnetic devices, combining strongly coupled electromagnetic-thermal-fluid-mechanical finite-element analysis, high-resolution lumped-parameter thermal networks, and operator-learning neural surrogates. I have authored **22 journal papers (17 in IEEE Transactions)**, **17 peer-reviewed conference papers**, and hold **9 patents**, recognised by the **2025 IEEE TEC Electric Vehicle Design Competition (1st Prize)**, the **2025 IEEE Transactions on Transportation Electrification Prize Paper Award (2nd Prize)**, and the **IEEE IAS CMD Thesis Contest (1st Prize, non-PhD)**. While my application domain has been high power-density electric drives, the underlying methodology is **application-agnostic** and well placed to support cross-disciplinary collaboration with the School's imaging and sensing groups, in particular on electromagnetic device design for **MRI gradient and shimming systems** and on **learned surrogates for inverse problems in electromagnetic imaging**.

A distinguishing feature of my profile is **full industrial development experience**. In partnership with **Inovance**, a leading EV traction-machine supplier, I have analysed, designed, manufactured and AVL-dynamometer-tested two advanced oil-cooled PM traction machines: an **18 000 rpm, 113 kW A0-class machine** (138 Nm peak torque, **7.78 kW/kg**, **95.3% peak efficiency**) and a **20 000 rpm, 270 kW A-class machine** (380 Nm peak torque, **7.87 kW/kg**, **97.4% peak efficiency**). I have also developed an **in-house finite-element code package** covering electromagnetic, structural-modal and mechanical-stress analysis of electrical machines, now being applied to magnet-free synchronous machine work at PEMC. This combination of academic and industrial credibility positions me well to pursue **ARC Discovery, DECRA and Linkage funding** and direct industrial partnerships once based at UQ.

I am prepared to teach into several of the listed curriculum areas, in particular **introductory programming for engineering students**, **intermediate systems programming**, **embedded and digital systems**, and **electrical engineering team and individual capstone projects**. My day-to-day toolchain spans **Python, MATLAB/Simulink, C/C++ for embedded platforms (TI C2000, STM32, FPGA), and machine-learning frameworks**, complemented by power-electronics prototyping (a three-phase full-bridge converter and its control built for testing flux-modulation machines) and an in-house Rust-based MCU data-acquisition tool. At postgraduate level I served as **teaching assistant for the SEU course Integrated Practice of Electrical Machine Systems**, preparing tutorial slides, running daily Q&A, reviewing student final-project outline proposals, and providing mark recommendations to the course coordinator. I am also **Co-Organiser and Co-Chair of a Special Session at ICEM 2026 (Covilhã, Portugal)** on *Next-Generation Electrified Transport*, jointly with Dr Lingyun Shao (NUAA).

Attached are my CV and detailed responses to the "About You" selection criteria. I would be very pleased to discuss how I could contribute to the School's teaching programme and research strategy.

Yours sincerely,

Wentao Zhang